

 HACKATHON PROJECT

Link Intel Suite

Internal Linking Intelligence, Powered by AI Agents

Drop in a Screaming Frog export or enter a URL → get a complete health score, topical clusters, orphan pages, PageRank analysis & contextual link recommendations.

5 AI Agents

< 60s Analysis

100% Local

Notion Export

THE PROBLEM

Internal linking is broken for most websites

SEO teams waste hours in spreadsheets trying to manually analyse link graphs that should be automated.

- Orphan pages get zero link equity — invisible to Google
- Generic anchors ('click here') dilute topical authority
- Broken internal links silently destroy crawl budget
- Keyword cannibalization tanks rankings without warning
- No visibility into PageRank flow across the site

The real cost

68%

of pages have
zero internal links

3–5h

wasted per audit
on manual analysis

40%

SEO impact from
internal link structure

THE SOLUTION

One tool. Complete internal link intelligence.



Built-in Crawler

Just enter a URL. No Screaming Frog needed.



Health Score

0–100 grade with weighted breakdown of every issue.



AI Agents

5 specialised agents run sequentially in under 60 seconds.



Link Recs

Contextual suggestions: which page should link to which, and what anchor to use.



Notion Export

Push a fully structured report to your Notion workspace in one click.



Slack Alerts

Get notified after every run with score, grade, and key stats.

5 AI Agents. One Pipeline.

Each agent is a specialised Python function. They run sequentially, each feeding results to the next.

01



Graph Agent

Builds the full internal link graph using NetworkX. Detects orphans, calculates PageRank, finds broken links and redirect chains.

02



Anchor Agent

Classifies every anchor text as descriptive, generic, empty, or over-optimised. Flags diversity issues per destination page.

03



Topic Agent

Clusters pages by TF-IDF cosine similarity into topical silos. Labels each cluster and identifies hub vs. scattered pages.

04



Entity Agent

Optionally calls Claude API to extract named entities and compute semantic relatedness between pages — enables smarter link suggestions.

05



Reporter

Computes the 0–100 health score, generates priority fix queue, writes report.json & report.html, pushes to Notion, and notifies Slack.

Everything in one tool

Built-in Crawler

Enter any URL — crawls up to 200 pages, robots.txt compliant. No extra tools needed.

CSV Upload

Drag & drop Screaming Frog exports for instant analysis on sites of any size.

Health Score 0–100

Weighted across 6 dimensions: broken links, orphans, anchors, PageRank, clusters, depth.

Orphan Detection

Finds zero-inlink pages invisible to Google, suggests which pages should link to them.

Anchor Analysis

Classifies every anchor (generic, empty, over-optimised, descriptive) per destination.

Topic Clusters

TF-IDF cosine similarity silos pages. Labels hubs vs. scattered with authority scores.

PageRank Flow

Simulates Google's PageRank — shows which pages gain or are starved of link equity.

Link Recommendations

Per-page suggestions with source, target, anchor text, and the reason why.

Cannibalization

Detects pages competing for the same keywords via title/H1 and anchor overlap.

Notion Export

One-click: pushes a fully structured report — score, fix checklist, recs, clusters.

Slack Alerts

Webhook sends grade, score, and key stats to your channel after every run.

History & Trends

Auto-saves 50 runs. Dashboard sparkline shows how your score changes over time.

LIVE DEMO

Demo walkthrough

Follow this order — each step builds on the previous one.

1

Open the Landing Page

localhost:7700/landing.html — show agent explainer & Notion section.

⌚ 30s

2

Enter a URL to Crawl

Crawl tab → enter domain → hit Analyse. Show live SSE feed.

⌚ 60s

3

Walk the Dashboard

Health Score gauge → Graph (orphans/broken) → Anchors → Clusters.

⌚ 90s

4

Open Recommendations

Link Recs page — source, target, anchor text, reason per page.

⌚ 30s

5

Export to Notion

Reports → Notion card → Export. Switch tab to show live result.

⌚ 30s

6

Show History Chart

Overview → History sparkline. Score trend across runs.

⌚ 20s

KEY TALKING POINTS



Built entirely with Python stdlib + Claude API — no paid crawl tools



Full analysis in under 60 seconds for a 200-page site



100% local — your data never leaves your machine



5 agents each do one job well: graph → anchor → topic → entity → report



Health score is objective & reproducible — not a black box



Plugs directly into existing workflows: Notion + Slack



Trend tracking: watch your score improve as you fix issues



Works on any site — no Screaming Frog licence required

COMPARISON

Why not just use existing tools?

Feature	Link Intel Suite	Screaming Frog	Ahrefs / Semrush	Custom Script
Built-in crawler	✓ Zero setup	✗ Separate app	✗	✗
Topical cluster detection	✓ AI-powered	✗	~ Basic	✗
PageRank simulation	✓ Full graph	✗	✗	~ DIY
Contextual link recs	✓ Per page	✗	~ Generic	✗
Notion export	✓ One click	✗	✗	✗
Health score history	✓ Auto-tracked	✗	~ Paid only	✗
Runs locally / free	✓ Fully local	~ £149/yr	✗ Subscription	✓
Setup time	< 2 minutes	30-60 min	Hours	Days

Built with the right tools

BACKEND

- Python 3 stdlib only — no frameworks
- ThreadingHTTPServer + SSE for live dashboard
- BFS web crawler (urllib + html.parser)
- NetworkX for link graph & PageRank

AI / ANALYSIS

- Claude API (claude-sonnet-4-6) for entity extraction
- TF-IDF cosine similarity for topical clusters
- Custom health scoring algorithm
- Anchor text classification logic

FRONTEND

- Vanilla JS SPA with hash routing
- Chart.js for data visualisation
- Server-Sent Events for live progress
- Dark theme with CSS custom properties

INTEGRATIONS

- Notion REST API v2022-06-28 (urllib)
- Slack Incoming Webhooks
- Screaming Frog CSV import
- JSON-based history persistence

IMPACT

What you get after one run

0–100

Health Score

< 60s

Full analysis

50+

Run history saved

5

AI Agents

WHAT THE REPORT INCLUDES



Priority fix queue — ranked by severity with page-level detail



Orphan pages with suggested linking pages



Topical cluster map — hub pages, scattered pages, gaps



Contextual link recommendations with suggested anchors & reasoning



Broken internal links list with source, destination & status



Anchor text breakdown — generic, empty, over-optimised per page



PageRank top 10 / bottom 10 with protection scores



Notion page + Slack notification — ready to share with the team

Try it now.

localhost:7700

Drop in any URL or Screaming Frog export and run a complete internal link audit in under 60 seconds.



No Screaming Frog required



Push to Notion in one click



Slack alerts out of the box



Health score + trend tracking